Lab 14.4 – Named Entity Recognition (NER)

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Install Required Libraries

```
!pip install spacy transformers torch pandas -q
!python -m spacy download en_core_web_sm
import spacy                      # NLP library for NER
from spacy import displacy        # For visualization
from transformers import pipeline # Hugging Face pipeline
import pandas as pd               # For DataFrame
import torch                      # For model backend
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_c
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 110.3 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

Load spaCy Model

```
nlp = spacy.load("en_core_web_sm")

print("spaCy pipeline components:")
print(nlp.pipe_names)
```

```
spaCy pipeline components:
['tok2vec', 'tagger', 'parser', 'attribute_ruler', 'lemmatizer', 'ner']
```

Prepare Sample Sentences

```
sentences = [
    "Apple launched the new iPhone 15 in California.",
    "Elon Musk is the CEO of Tesla and SpaceX.",
    "India won the cricket match against Australia in Mumbai.",
    "Google announced a partnership with Microsoft in New York.",
    "Narendra Modi met Joe Biden at the White House."
]

sentences
```

```
['Apple launched the new iPhone 15 in California.',
 'Elon Musk is the CEO of Tesla and SpaceX.',
 'India won the cricket match against Australia in Mumbai.',
 'Google announced a partnership with Microsoft in New York.',
 'Narendra Modi met Joe Biden at the White House.']
```

Apply spaCy NER

```
spacy_results = []

for sentence in sentences:
    doc = nlp(sentence)

    print(f"\nSentence: {sentence}")
    for ent in doc.ents:
        print(f"Entity: {ent.text} | Label: {ent.label_}")

        spacy_results.append({
            "Sentence": sentence,
            "Entity": ent.text,
            "Label": ent.label_
        })
```

```
Sentence: Apple launched the new iPhone 15 in California.
Entity: Apple | Label: ORG
Entity: 15 | Label: CARDINAL
Entity: California | Label: GPE

Sentence: Elon Musk is the CEO of Tesla and SpaceX.
Entity: Elon Musk | Label: PERSON
Entity: Tesla | Label: ORG

Sentence: India won the cricket match against Australia in Mumbai.
Entity: India | Label: GPE
Entity: Australia | Label: GPE
Entity: Mumbai | Label: GPE

Sentence: Google announced a partnership with Microsoft in New York.
Entity: Google | Label: ORG
Entity: Microsoft | Label: ORG
Entity: New York | Label: GPE

Sentence: Narendra Modi met Joe Biden at the White House.
Entity: Narendra Modi | Label: PERSON
Entity: Joe Biden | Label: PERSON
Entity: the White House | Label: FAC
```

Store spaCy Results in DataFrame

```
spacy_df = pd.DataFrame(spacy_results)
spacy_df
```

	Sentence	Entity	Label
0	Apple launched the new iPhone 15 in California.	Apple	ORG
1	Apple launched the new iPhone 15 in California.	15	CARDINAL
2	Apple launched the new iPhone 15 in California.	California	GPE
3	Elon Musk is the CEO of Tesla and SpaceX.	Elon Musk	PERSON
4	Elon Musk is the CEO of Tesla and SpaceX.	Tesla	ORG
5	India won the cricket match against Australia ...	India	GPE
6	India won the cricket match against Australia ...	Australia	GPE
7	India won the cricket match against Australia ...	Mumbai	GPE
8	Google announced a partnership with Microsoft ...	Google	ORG
9	Google announced a partnership with Microsoft ...	Microsoft	ORG
10	Google announced a partnership with Microsoft ...	New York	GPE
11	Narendra Modi met Joe Biden at the White House.	Narendra Modi	PERSON
12	Narendra Modi met Joe Biden at the White House.	Joe Biden	PERSON
13	Narendra Modi met Joe Biden at the White House.	the White House	FAC

Next steps: Generate code with spacy_df New interactive sheet

Visualize spaCy NER

```
for sentence in sentences:
    doc = nlp(sentence)
    displacy.render(doc, style="ent", jupyter=True)
```

Apple **ORG** launched the new iPhone 15 **CARDINAL** in California **GPE** .

Elon Musk **PERSON** is the CEO of Tesla **ORG** and SpaceX.

India **GPE** won the cricket match against Australia **GPE** in Mumbai **GPE** .

Google **ORG** announced a partnership with Microsoft **ORG** in New York **GPE** .

Narendra Modi **PERSON** met Joe Biden **PERSON** at the White House **FAC** .

Load Hugging Face NER Model

```
hf_ner = pipeline(
    "ner",
    model="dslim/bert-base-NER"
)
```

Loading weights: 100% 199/199 [00:00<00:00, 829.94it/s, Materializing param=classifier.weight]

```
BertForTokenClassification LOAD REPORT from: dslim/bert-base-NER
Key                      | Status     |  |
-------------------------+------------+--+-
bert.pooler.dense.bias   | UNEXPECTED |  |
bert.pooler.dense.weight | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expec
```

Apply Transformer NER

Gemini

```
hf_results = []

for sentence in sentences:
    ner_results = hf_ner(sentence)

    print(f"\nSentence: {sentence}")
    for entity in ner_results:
        print(f"Entity: {entity['word']} | Label: {entity['entity']} | Score: {entity['score']:.2

        hf_results.append({
            "Sentence": sentence,
            "Entity": entity['word'],
            "Label": entity['entity'],
            "Score": entity['score']
        })

# Create a new cell to store Hugging Face results in a DataFrame
# Adding a new cell to store results.
# Create a new cell to store Hugging Face results in a DataFrame
hf_df = pd.DataFrame(hf_results)
hf_df
```

```
Sentence: Apple launched the new iPhone 15 in California.
Entity: Apple | Label: B-ORG | Score: 1.00
Entity: iPhone | Label: B-MISC | Score: 1.00
Entity: 15 | Label: I-MISC | Score: 1.00
Entity: California | Label: B-LOC | Score: 1.00

Sentence: Elon Musk is the CEO of Tesla and SpaceX.
Entity: El | Label: B-PER | Score: 0.99
Entity: ##on | Label: B-PER | Score: 0.84
Entity: Mu | Label: I-PER | Score: 1.00
Entity: ##sk | Label: I-PER | Score: 0.97
Entity: Te | Label: B-ORG | Score: 1.00
Entity: ##sla | Label: I-ORG | Score: 1.00
Entity: Space | Label: B-ORG | Score: 1.00
Entity: ##X | Label: I-ORG | Score: 1.00

Sentence: India won the cricket match against Australia in Mumbai.
Entity: India | Label: B-LOC | Score: 1.00
Entity: Australia | Label: B-LOC | Score: 1.00
Entity: Mumbai | Label: B-LOC | Score: 1.00

Sentence: Google announced a partnership with Microsoft in New York.
Entity: Google | Label: B-ORG | Score: 1.00
Entity: Microsoft | Label: B-ORG | Score: 1.00
Entity: New | Label: B-LOC | Score: 1.00
Entity: York | Label: I-LOC | Score: 1.00

Sentence: Narendra Modi met Joe Biden at the White House.
Entity: Na | Label: B-PER | Score: 1.00
Entity: ##ren | Label: B-PER | Score: 0.81
Entity: ##dra | Label: B-PER | Score: 0.76
Entity: Mo | Label: I-PER | Score: 1.00
Entity: ##di | Label: I-PER | Score: 0.92
Entity: Joe | Label: B-PER | Score: 1.00
Entity: B | Label: I-PER | Score: 1.00
Entity: ##iden | Label: I-PER | Score: 0.95
Entity: White | Label: B-LOC | Score: 1.00
Entity: House | Label: I-LOC | Score: 1.00
```

Store Transformer Results in DataFrame

```
hf_df = pd.DataFrame(hf_results)
hf_df
```

	Sentence	Entity	Label	Score
0	Apple launched the new iPhone 15 in California.	Apple	B-ORG	0.997462
1	Apple launched the new iPhone 15 in California.	iPhone	B-MISC	0.998376
2	Apple launched the new iPhone 15 in California.	15	I-MISC	0.998245
3	Apple launched the new iPhone 15 in California.	California	B-LOC	0.999744
4	Elon Musk is the CEO of Tesla and SpaceX.	El	B-PER	0.992953
5	Elon Musk is the CEO of Tesla and SpaceX.	##on	B-PER	0.835798
6	Elon Musk is the CEO of Tesla and SpaceX.	Mu	I-PER	0.998362
7	Elon Musk is the CEO of Tesla and SpaceX.	##sk	I-PER	0.970736
8	Elon Musk is the CEO of Tesla and SpaceX.	Te	B-ORG	0.999043
9	Elon Musk is the CEO of Tesla and SpaceX.	##sla	I-ORG	0.996143
10	Elon Musk is the CEO of Tesla and SpaceX.	Space	B-ORG	0.998576
11	Elon Musk is the CEO of Tesla and SpaceX.	##X	I-ORG	0.999073
12	India won the cricket match against Australia ...	India	B-LOC	0.999816
13	India won the cricket match against Australia ...	Australia	B-LOC	0.999799
14	India won the cricket match against Australia ...	Mumbai	B-LOC	0.999503
15	Google announced a partnership with Microsoft ...	Google	B-ORG	0.998756
16	Google announced a partnership with Microsoft ...	Microsoft	B-ORG	0.998762
17	Google announced a partnership with Microsoft ...	New	B-LOC	0.999594
18	Google announced a partnership with Microsoft ...	York	I-LOC	0.999472
19	Narendra Modi met Joe Biden at the White House.	Na	B-PER	0.999327
20	Narendra Modi met Joe Biden at the White House.	##ren	B-PER	0.810916
21	Narendra Modi met Joe Biden at the White House.	##dra	B-PER	0.758017
22	Narendra Modi met Joe Biden at the White House.	Mo	I-PER	0.999561
23	Narendra Modi met Joe Biden at the White House.	##di	I-PER	0.921693
24	Narendra Modi met Joe Biden at the White House.	Joe	B-PER	0.999739
25	Narendra Modi met Joe Biden at the White House.	B	I-PER	0.999621
26	Narendra Modi met Joe Biden at the White House.	##iden	I-PER	0.952399
27	Narendra Modi met Joe Biden at the White House.	White	B-LOC	0.999348
28	Narendra Modi met Joe Biden at the White House.	House	I-LOC	0.998806

Next steps: Generate code with hf_df New interactive sheet

Comparison Table

```
comparison = pd.DataFrame({
    "Feature": [
        "Model Type",
        "Speed",
        "Context Handling",
        "Confidence Score",
```

```
            "Computational Cost"
        ],
        "spaCy": [
            "Statistical + Rule-based",
            "Very Fast",
            "Moderate",
            "No",
            "Low"
        ],
        "Hugging Face": [
            "Transformer (BERT)",
            "Slower",
            "Very Strong (Bidirectional)",
            "Yes",
            "High"
        ]
    })

    comparison
```

	Feature	spaCy	Hugging Face
0	Model Type	Statistical + Rule-based	Transformer (BERT)
1	Speed	Very Fast	Slower
2	Context Handling	Moderate	Very Strong (Bidirectional)
3	Confidence Score	No	Yes
4	Computational Cost	Low	High

Next steps: Generate code with `comparison` New interactive sheet